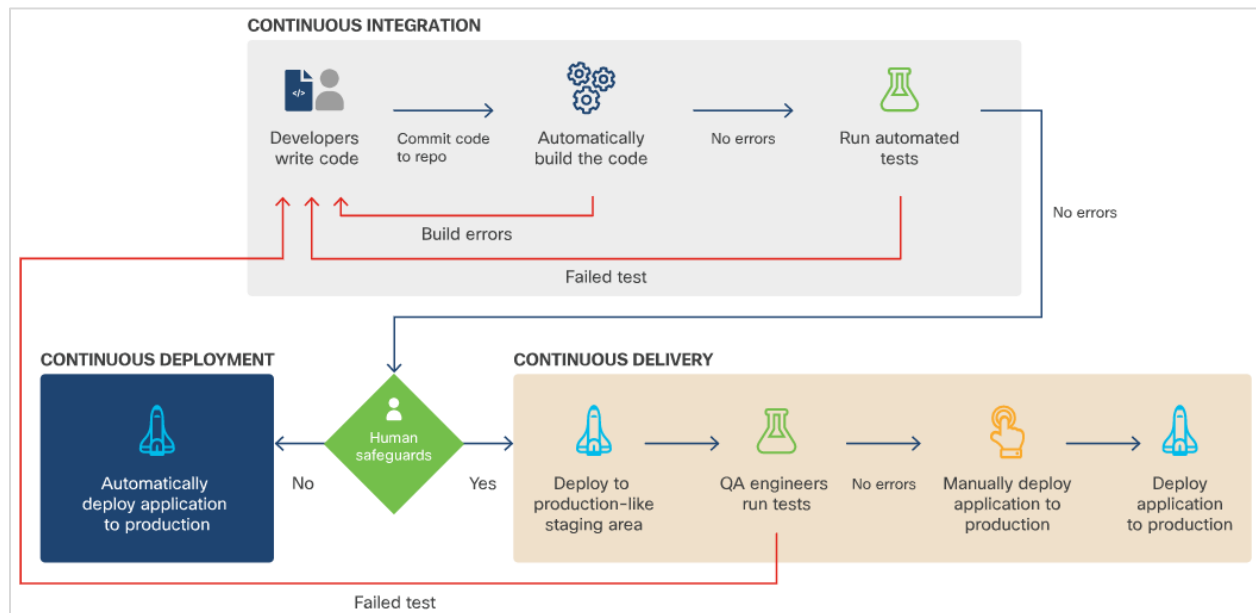


Continuous Integration (CI):

Continuous Integration (CI) is a development practice that requires developers to integrate code into a shared repository several times a day. Each check-in is then verified by an automated build, allowing teams to detect problems early. you need to run code integration multiple times a day, every day, and you need to run the automated verification of the integration. Therefore, your team can resolve the integration errors much faster. Continuous Integration doesn't get rid of bugs, but it does make them dramatically easier to find & remove. Continuous Integration enables the developers on the project to continually merge the changes with the main branch of the existing application.



Continuous Deployment (CD):

Continuous Deployment is closely related to Continuous Integration and refers to the release into production of software that passes the automated tests. Essentially, it is the practice of releasing every good build to users. Continuous deployment is about automating the release of a good build to the production environment. It is a special type of Continuous Delivery in which, every version of software that is marked as ready for production gets deployed.

Continuous Delivery (CD):

Continuous delivery (CD) is about ensuring that every good build is potentially ready for production release.

Continuous Deployment



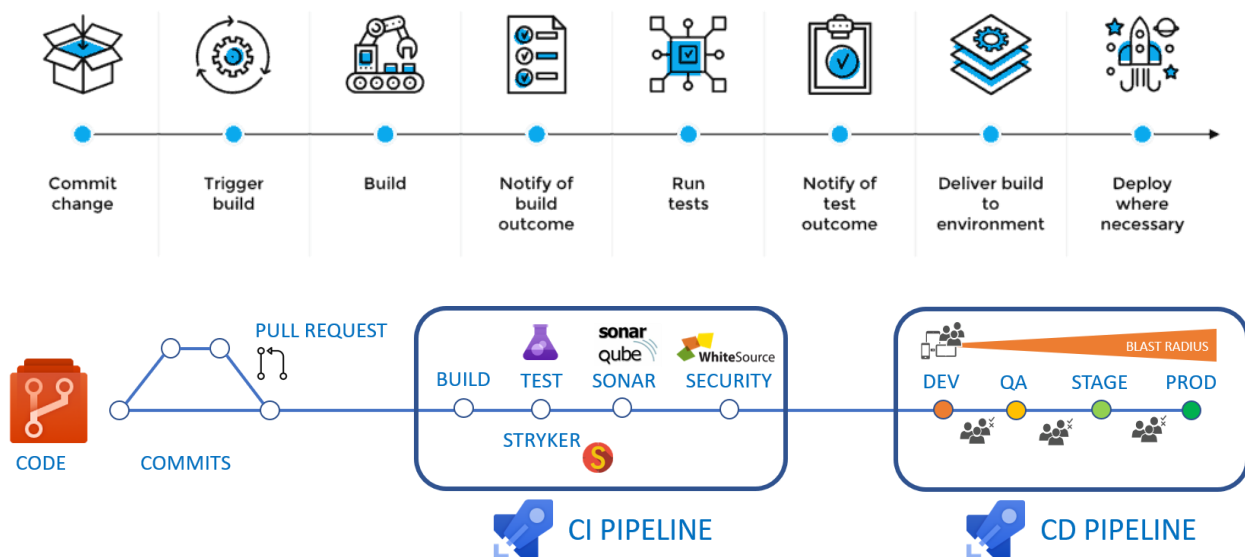
Continuous Delivery



CI/CD Pipeline:

First it was Waterfall model, next it was Agile, and now it's DevOps. With the rise of DevOps has come the new methods of Continuous Integration, Continuous Delivery, (CI/CD) and Continuous Deployment. Continuous Integration/Continuous Deployment (CI/CD) is a philosophy for software deployment. A CI/CD Pipeline can be easily understood as the process pathway through which we can deliver a single unit of production-ready software. You always need to juggle resources (developers, budget, time) within the constraints of the business (time to market, deadline, runway). As such, you need to put more wood behind fewer arrows for areas that give you the best bang for your buck. Reducing costs and complexity is a crucial benefit of employing a CI/CD pipeline.

CI/CD Pipeline



CI/CD:

CI stands for **Continuous Integration** and CD stands for **Continuous Delivery** and **Continuous Deployment**. You can think of it as a process which is similar to a software development lifecycle. With CI/CD, most software releases go through the set of stages, a failure in any stage typically triggers a notification. CI/CD bridges the gap between development and operations teams by automating build, test and deployment of applications. Software will move along the various phases or stages in this lifecycle, before it is delivered to the customer or before it is live on production. Our task is to automate the pipeline in order to make the entire software development lifecycle on the dev-ops mode/ automated mode. For this, they would need automation tools.

